Step 1 :- Install and load required packages

```
packages <- c("rvest", "httr", "xml2", "dplyr", "stringr", "tibble")

install.packages(setdiff(packages, rownames(installed.packages())))

library(rvest)
library(httr)
library(xml2)
library(dplyr)
library(stringr)
library(tibble)
```

(Fig 2.1) These packages help read webpages, parse HTML, clean text, organize data, and create tables.

- *rvest package used for web scraping*
- *httr package to send a request to the website.*
- *Xml2 package to read and parse HTML/XML content.*
- *Dplyr package used for data cleaning and organizing.*
- *Stringr package used for text cleaning.*
- *Tibble package used to create a clean table.*

Step 2 : - Set Amazon laptop search URL

```
url <- "https://www.amazon.in/s?k=laptop"
```

(Fig 2.2) Stores the Amazon laptop search URL. This tells R which webpage to extract data from.

Step 3 :- Set output folder path

```
output_folder <- "D:/SWAPNIL FOLDER/ADYPU/FY Notes ( Sem 7)/DAR/Assignment -1/amazon_laptop_ou
dir.create(output_folder, showWarnings = FALSE)
```

(Fig 2.3) The CSV file and saved HTML page will be stored in this folder.

Step 4 :- Read Amazon webpage

```
response <- GET(
  url,
  add_headers(
    "User-Agent" = "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 Chrome/120 Safari/537.36",
    "Accept-Language" = "en-IN,en;q=0.9",
    "Accept" = "text/html,application/xhtml+xml,application/xml;q=0.9,*/*;q=0.8"
  )
)

page <- read_html(content(response, "text", encoding = "UTF-8"))
```

(Fig 2.4) Sends a request to Amazon using browser-like headers. This helps R access the webpage like a normal browser.

Step 5 :- Select laptop product blocks

```
products <- page %>%
  html_elements("div[data-component-type='s-search-result']")
```

(Fig 2.5) Converts the webpage response into HTML text. It also saves the received page as an HTML file for checking.

Step 6: Stop if no products found

```
if (length(products) == 0) {
  stop("No product data found. Amazon may be blocking scraping.")
}
```

(Fig 2.6) This helps confirm whether Amazon opened correctly or showed a block page.

Step 7 :- Function to safely extract text

```
get_text <- function(product, selector) {
  value <- product %>%
    html_element(selector) %>%
    html_text2()

  ifelse(is.na(value), "Not Available", str_squish(value))
}
```

(Fig 2.7) Selects all laptop product blocks from the webpage. Each product block contains details about one laptop.

Step 8 :- Function to safely extract product URL

```
get_link <- function(product, selector) {
  value <- product %>%
    html_element(selector) %>%
    html_attr("href")

  ifelse(
    is.na(value),
    "Not Available",
    paste0("https://www.amazon.in", value)
  )
}
```

(Fig 2.8) Checks whether product blocks were found. If none are found, the program stops and shows a message.

Step 9 :- Extract laptop details

```
amazon_laptop_data <- tibble(
  laptop_name = sapply(products, get_text, "h2 span"),
  price = sapply(products, get_text, ".a-price .a-offscreen"),
  rating = sapply(products, get_text, ".a-icon-alt"),
  reviews = sapply(products, get_text, "a[href*='customerReviews'] span"),
  delivery_date = sapply(products, get_text, "[data-cy='delivery-recipe']"),
  product_url = sapply(products, get_link, "h2 a")
)
```

(Fig 2.9) Creates a helper function to safely extract text. If a value is missing, it returns Not Available.

Step 10 :- Clean the data

```
amazon_laptop_data <- amazon_laptop_data %>%
  filter(laptop_name != "Not Available") %>%
  mutate(
    price = ifelse(price == "", "Not Available", price),
    rating = ifelse(rating == "", "Not Available", rating),
    reviews = ifelse(reviews == "", "Not Available", reviews),
    delivery_date = ifelse(delivery_date == "", "Not Available", delivery_date),
    across(everything(), str_squish)
  )
```

(Fig 2.10) Creates a helper function to safely extract product links. It converts relative links into full Amazon URLs.

Step 11 :- Stop if extracted table is empty

```
if (nrow(amazon_laptop_data) == 0) {
  stop("No laptop data extracted. Amazon blocked the request or changed page structure.")
}
```

(Fig 2.11) Extracts laptop name, price, rating, reviews, delivery date, and product URL from each product block.

Step 12 :- Save data into CSV file

```
csv_path <- file.path(output_folder, "amazon_laptop_data.csv")

write.csv(amazon_laptop_data, csv_path, row.names = FALSE)
```

(Fig 2.12) The cleaned laptop data is saved as a CSV file. The file name is amazon_laptop_data.csv.

Step 13 :- Print and view final data

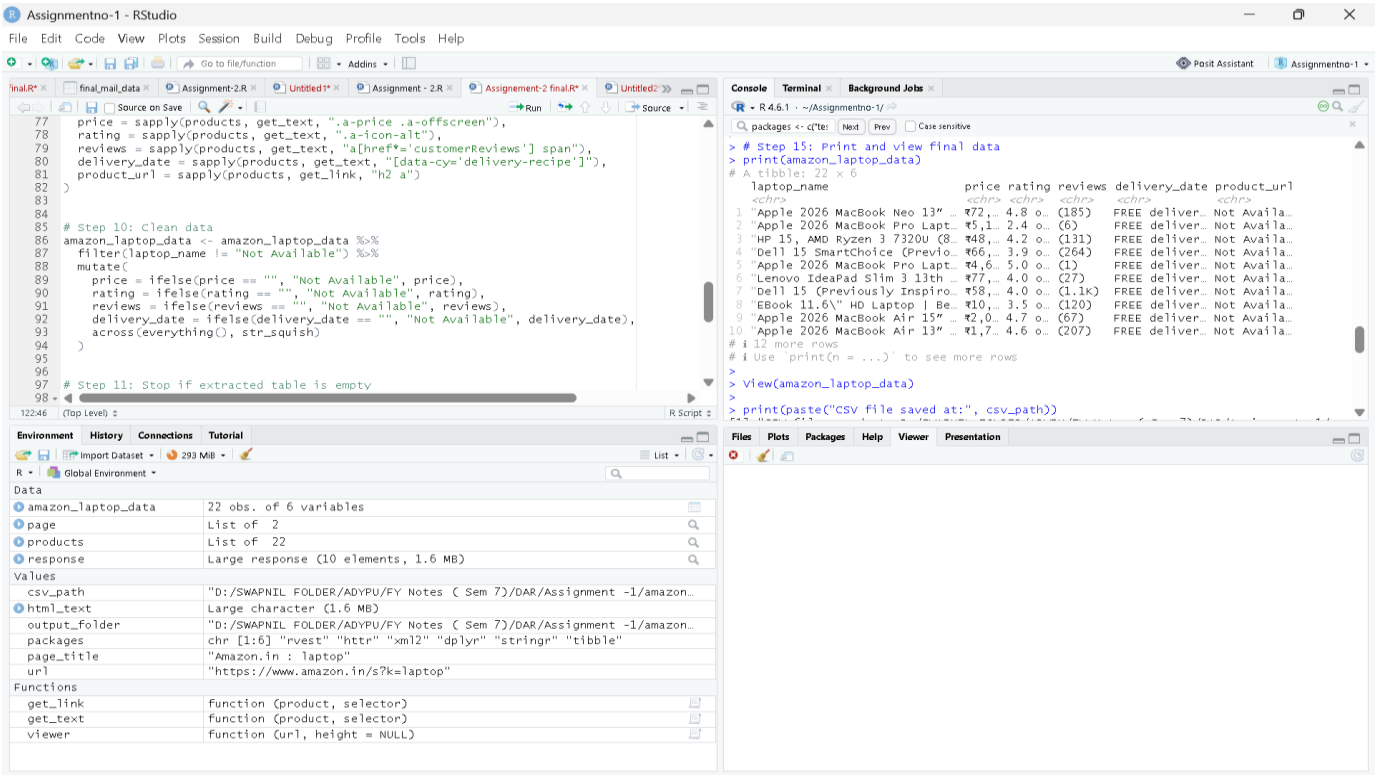
```
print(amazon_laptop_data)

View(amazon_laptop_data)

print(paste("CSV file saved at:", csv_path))
```

(Fig 2.13) The final extracted data is printed and opened in RStudio. The CSV file location is also displayed.

R Studio Full Page Displayed



(Fig 2.14) R Studio Console Page

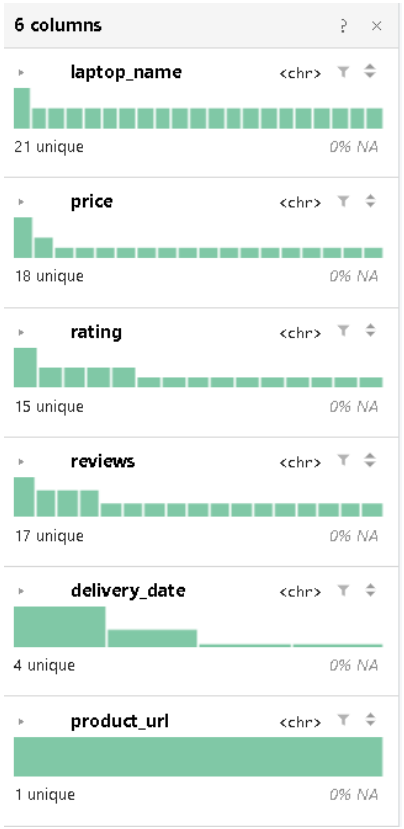
Final Output

	laptop_name	price	rating	reviews
1	Apple 2026 MacBook Neo 13" Laptop with A18 Pro chip: Built for AI and Apple Intelligence, Liqui...	₹72,990	4.8 out of 5 stars	(185)
2	Apple 2026 MacBook Pro Laptop with M5 Max chip with 18-core CPU and 32-core GPU: Built for A...	₹5,13,990	2.4 out of 5 stars	(6)
3	HP 15, AMD Ryzen 3 7320U (8GB LPDDR5, 512GB SSD) FHD, Anti-Glare, Micro-Edge, 15.6"/39.6cm...	₹48,990	4.2 out of 5 stars	(131)
4	Dell 15 SmartChoice (Previously Inspiron), Intel 13th Gen Core i5-1334U, 16GB, 1TB SSD, FHD,15.6"...	₹66,990	3.9 out of 5 stars	(264)
5	Apple 2026 MacBook Pro Laptop with M5 Max chip with 18-core CPU and 32-core GPU: Built for A...	₹4,64,490	5.0 out of 5 stars	(1)
6	Lenovo IdeaPad Slim 3 13th Gen Core i7-13620H 15.3" (38.8cm) WUXGA IPS Laptop (16GB RAM/5...	₹77,990	4.0 out of 5 stars	(27)
7	Dell 15 (Previously Inspiron) Laptop, 14th Gen Intel Core i3/Core 3 100U Processor, 16GB DDR4, 51...	₹58,990	4.0 out of 5 stars	(1.1K)
8	EBook 11.6" HD Laptop	Best Student & Office Work Laptop	Celeron N4020	4GB DDR4	128GB...	₹10,990	3.5 out of 5 stars	(120)
9	Apple 2026 MacBook Air 15" Laptop with M5 chip: AI and Apple Intelligence, 38.91 cm (15.3") Liq...	₹2,00,490	4.7 out of 5 stars	(67)
10	Apple 2026 MacBook Air 13" Laptop with M5 chip: AI and Apple Intelligence, 34.46 cm (13.6") Liq...	₹1,72,490	4.6 out of 5 stars	(207)
11	Acer Smartchoice Aspire One, AMD Ryzen 3-7320U, Office 2024 + M365 Basic, 8GB LPDDR5 RAM/...	₹37,990	3.2 out of 5 stars	(6)
12	ASUS Vivobook 15, Smartchoice,Intel Core i5 13th Gen 13420H,16GB RAM, 512GB SSD, FHD 15.6",...	₹63,990	4.1 out of 5 stars	(443)
13	Dell 15 (Previously Inspiron) Laptop, 14th Gen Intel Core i3/Core 3 100U Processor, 8GB DDR4, 512...	₹51,490	3.4 out of 5 stars	(11)
14	Acer Chromebook, MediaTek Kompanio 540 Octa-Core Processor, 4GB RAM, 64GB eMMC, Full HD,...	₹19,990	5.0 out of 5 stars	(5)
15	ASUS TUF A15, AMD Ryzen 7 170, RTX 3050-4GB, 16GB RAM (Upgradeable), 512GB SSD, FHD, 15....	₹70,990	4.2 out of 5 stars	(27)
16	HP OmniBook 5 OLED (Previously Pavilion), Snapdragon X Processor 45 Tops (16GB LPDDR5x, 512...	₹75,990	4.2 out of 5 stars	(120)
17	Acer Smartchoice Aspire One, AMD Ryzen 5-40, 8GB LPDDR5 RAM/ 512GB SSD, 14.0"/35.56cm W...	₹43,990	4.2 out of 5 stars	(6)
18	Apple 2026 MacBook Neo 13" Laptop with A18 Pro chip: Built for AI and Apple Intelligence, Liqui...	₹72,990	4.8 out of 5 stars	(185)
19	HP Victus, 13th Gen Intel Core i5-13420H, 4GB RTX 2050, 16GB DDR4 (Upgradeable) 512GB SSD, 1...	₹72,990	3.8 out of 5 stars	(92)
20	ASUS Vivobook 16, Smartchoice, Intel Core Ultra 5 Series 2, 16GB RAM, 512GB SSD, FHD+ 16", Wi...	₹77,990	4.1 out of 5 stars	(233)
21	Lenovo V14 Intel Core i3 13th Gen 14" FHD (1920x1080) Antiglare 250 Nits Thin and Light Laptop (...	₹52,990	3.6 out of 5 stars	(9)
22	Apple 2026 MacBook Neo 13" Laptop with A18 Pro chip: Built for AI and Apple Intelligence, Liqui...	₹72,990	4.9 out of 5 stars	(37)

(Fig 2.15) Laptop name, Price, Rating and Reviews are displayed above.

delivery_date	product_url
FREE delivery Tue, 18 Aug Or fastest delivery Tomorrow 6 am - 10 am	Not Available
FREE delivery Tue, 18 Aug	Not Available
FREE delivery Tue, 18 Aug	Not Available
FREE delivery Tue, 18 Aug	Not Available
FREE delivery Tue, 18 Aug	Not Available
FREE delivery Tue, 18 Aug	Not Available
FREE delivery Tue, 18 Aug	Not Available
FREE delivery Tue, 18 Aug	Not Available
FREE delivery Tue, 18 Aug	Not Available
FREE delivery Tue, 18 Aug	Not Available
FREE delivery Tue, 18 Aug Or fastest delivery Tomorrow 6 am - 10 am	Not Available
FREE delivery Tue, 18 Aug	Not Available
FREE delivery Tue, 18 Aug	Not Available
FREE delivery Tue, 18 Aug Or fastest delivery Tomorrow 6 am - 10 am	Not Available
FREE delivery	Not Available
FREE delivery Tue, 18 Aug Or fastest delivery Tomorrow 6 am - 10 am	Not Available
FREE delivery Tue, 18 Aug	Not Available
FREE delivery Tue, 18 Aug Or fastest delivery Tomorrow 6 am - 10 am	Not Available
FREE delivery Tue, 18 Aug	Not Available
FREE delivery Wed, 19 Aug	Not Available
FREE delivery Tue, 18 Aug	Not Available
FREE delivery Tue, 18 Aug Or fastest delivery Tomorrow 6 am - 10 am	Not Available

(Fig 2.16) Delivery Date and Product Url which is currently not mentioned.



(Fig 2.17) Through Visual Graphs we can understand easily.